

OPERATIVE REPORT

Member ID:	MA-000004	DOB:	April 18, 1952
Patient Name:	Walter Pryor	Age:	71 years
Procedure Date:	June 10, 2024	Surgeon:	Dr. Michael J. Chen, MD

PRIMARY DIAGNOSIS

Primary:	Unilateral primary osteoarthritis, right knee (M17.11)
Severity:	Kellgren-Lawrence Grade IV (bone-on-bone), medial and lateral compartments
Indication:	End-stage OA with failed conservative management, suitable for total knee replacement

PROCEDURE

CPT Code:	27447
Description:	Arthroplasty, knee, condyle and plateau; medial AND lateral compartments with patella resurfacing (total knee arthroplasty)
Laterality:	Right knee (unilateral)

PREOPERATIVE IMAGING FINDINGS

Right Knee Radiographs (June 6, 2024): Kellgren-Lawrence Grade IV osteoarthritis with severe joint space narrowing bilaterally (medial and lateral compartments), bone-on-bone articulation, large osteophytes on femoral condyles and tibial plateau, subchondral sclerosis, and subchondral cyst formation. Findings are consistent with end-stage OA suitable for total knee arthroplasty. This Grade IV imaging directly supports the indication for major joint replacement rather than arthroscopic procedures.

DOCUMENTED CONSERVATIVE MANAGEMENT

Date Range	Treatment	Provider/Notes	Outcome
December 2023	Physical Therapy	8 visits, ROM and strengthening exercises	Minimal improvement
January 2024	NSAIDs	Ibuprofen 600mg TID, naproxen as needed	Mild symptom relief only
February 2024	Corticosteroid Injection #1	Intra-articular triamcinolone 40mg	Relief lasted 4 weeks
April 2024	Corticosteroid Injection #2	Intra-articular triamcinolone 40mg	Relief lasted 2-3 weeks
May 2024	PT resumed	4 additional visits	No functional improvement

Summary: Patient has documented conservative management spanning 6+ months (December 2023–May 2024) including physical therapy (12 visits), NSAIDs, and two intra-articular corticosteroid injections. All conservative measures failed to provide lasting relief. This meets CMS LCD L33518 requirements for TKA candidacy.

OPERATIVE FINDINGS

Patient was positioned supine on operative table with right knee slightly flexed in a knee holder. After appropriate anesthesia and sterile preparation and draping, the right knee was prepared in standard surgical fashion. **Direct visualization and assessment confirmed:**

- Medial compartment: Full-thickness cartilage loss (bone-on-bone), degenerative changes on femoral condyle and tibial plateau
- Lateral compartment: Advanced cartilage degeneration with bone exposure, moderate osteophyte burden
- Patellofemoral compartment: Grade III-IV chondral damage, anterior knee pain on palpation
- Synovium: Proliferative, consistent with chronic inflammatory osteoarthritis

- Ligaments: ACL/PCL intact, collateral ligaments stable
- Menisci: Degenerative tears consistent with severe OA, no significant loose bodies Findings confirm end-stage tricompartmental osteoarthritis suitable for total knee arthroplasty.

SURGICAL TECHNIQUE

Standard medial parapatellar approach was used. Quadriceps tendon was divided to provide exposure. Distal femoral and proximal tibial cuts were made using a modular cutting guide system (Stryker Triathlon, size 4.5 femur / size 4.0 tibia) with appropriate varus/valgus and flexion/extension alignment using intramedullary alignment guides. All osteophytes were removed. Soft tissue balancing was performed to achieve neutral ligamentous balance in extension and flexion (gap balancing technique). Fixed-bearing prosthesis (Stryker Triathlon Cruciate Retaining) was selected. Patellar resurfacing (all-polyethylene component) was performed given advanced patellar cartilage loss. Components were cemented in place with antibiotic-loaded cement (Palacos with gentamicin and vancomycin). Final reduction was verified; knee achieved full extension and 110° flexion. No ligamentous instability. Alignment verified within acceptable parameters (Mechanical axis within 3° varus/valgus). Hemostasis was achieved with electrocautery. Quadriceps tendon was repaired with #2 Ethibond sutures. Vastus medialis obliquus was repaired. Arthrotomy was closed with #1 Vicryl in running fashion. Subcutaneous tissues and skin were closed in layers with skin staples. Sterile dressing and compression wrap applied.

INTRAOPERATIVE DATA

Estimated Blood Loss:	350 mL
Operative Time:	75 minutes
Tourniquet Time:	65 minutes
Intraoperative Complications:	None
Drain Placed:	Yes, hemovac drain, 0.5L output

PROSTHETIC IMPLANTS USED

Prosthesis:	Stryker Triathlon Cruciate Retaining TKA
Femoral Component:	Size 4.5, 3° valgus
Tibial Component:	Size 4.0, modular polyethylene insert, standard conformity
Patellar Component:	Size 4.0, all-polyethylene
Fixation:	Cemented (Palacos with antibiotic)

CLINICAL SUMMARY

This operative report documents a complete total knee arthroplasty (CPT 27447) with both medial and lateral compartment replacement and patellar resurfacing for end-stage primary osteoarthritis of the right knee. The procedure was performed on June 10, 2024 on a patient with Kellgren-Lawrence Grade IV radiographic changes, documented 6+ months of failed conservative management, and clear indication for major joint replacement. All elements required by CMS LCD L33518 are present and documented: preoperative imaging at Grade IV severity, documented conservative management trial, operative report with surgical detail, and appropriate prosthetic selection. This claim is medically necessary and appropriately coded.